

RISHAB MAHATO

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SUMMARY

A Software Engineering student with strong problem-solving skills and hands-on experience in machine learning, data science, and web development. Proficient in building end-to-end ML pipelines, deriving actionable insights from data, and developing scalable full-stack applications. Solid understanding of data structures, algorithms, and modern AI tools.

EDUCATION

Parul Institute of Engineering and Technology

Bachelor of Technology, Computer Science (Specialisation in Artificial Intelligence)

2023 – 2027

CGPA: 8.3/10

INTERNSHIP EXPERIENCE

R&D Intern – Shritu Technology Pvt Ltd.

(May 2026 – Present):

Conducting research on AI technologies and identifying the best implementation strategies for Oracle Integration Cloud (OIC) and Oracle Fusion applications. Analyzing AI-driven automation solutions, optimizing workflows, and exploring innovative approaches to improve enterprise integration and business processes.

PROJECTS

AI-Based Voice Interview System with Resume Analysis | MERN, NLP, STT/TTS | [GitHub](#)

- Built a **real-time AI-powered interview system** using MERN stack with resume parsing and dynamic question generation.
- Implemented **NLP-based resume analysis** to extract skills and tailor interview questions based on candidate profile.
- Integrated **speech processing (STT/TTS)** for seamless voice interaction between user and system.
- Developed an **automated evaluation system** using NLP techniques to score responses and provide real-time feedback.

Smart Analytics Dashboard | Python, Pandas, Scikit-learn, XGBoost, Full Stack | [GitHub](#) | [Live Demo](#)

- Designed and implemented an **end-to-end ML pipeline** for data preprocessing, feature engineering, and model training.
- Achieved **~91% accuracy (R^2)** in sales prediction using Scikit-learn and XGBoost models.
- Built a **full-stack analytics dashboard** to visualize trends, predictions, and model performance in real time.
- Developed backend APIs for **data integration, model inference, and feature visualization** to support scalable insights.

QuerySafe — AI-Powered Database Assistant | Python, FastAPI, Next.js, LangChain, Docker | [GitHub](#)

- Built an **AI-powered read-only database query system** using LangChain & GPT-4/Claude, enabling non-technical teams to query databases in plain English without writing SQL.
- Implemented a **3-layer SQL safety engine** — prompt-level restriction, sqlparse AST validation, and read-only DB driver enforcement — blocking all destructive operations (INSERT, UPDATE, DELETE, DROP).
- Designed **role-based access control (RBAC)** with AES-256 credential encryption, JWT + TOTP two-factor authentication, and full audit logging for enterprise-grade security.
- Deployed a **multi-database adapter system** (PostgreSQL, MySQL, MongoDB, Snowflake, BigQuery) with ChromaDB vector store for schema-aware prompting and Redis session memory.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript

AI/ML / Data Science: Scikit-learn, XGBoost, PyTorch, LangChain, NLP, Transformers, Pandas, NumPy, Matplotlib

Web Frameworks: FastAPI, Django, Flask, Node.js, Express.js, React.js, Next.js, Bootstrap

Databases: PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, Snowflake

Cloud & DevOps: Docker, Docker Compose, Nginx, GitHub Actions, Git, Linux

Developer Tools: VS Code, Jupyter Notebook, Postman, Swagger

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Machine Learning, Deep Learning, Dynamic Programming

CERTIFICATIONS

NPTEL Certification – Computer Networks and Internet Protocol – Completed an in-depth course on computer networking concepts including TCP/IP, routing, congestion control, and application layer protocols. ([Certificate](#))

Oracle Cloud Infrastructure (OCI) – Cloud services, computing, storage, networking, and deployment on Oracle's cloud platform. ([certification](#))

Web Development – Front-end (HTML, CSS, JavaScript) and back-end development with databases (SQL, MongoDB).